

Sri Lanka Institute of Information Technology



IT 1040 - Fundamentals of Computing

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Water Tank Monitoring System

Proposal Document

Project Group No: P13-12

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Water Tank Monitoring System Project Proposal



Faculty of Computing

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Background

- Water is one of the most essential solvent which plays an important role in everyday life. Efficient management of water resources is of prime importance in areas that experience less rain or inequitable rainfall.
- Water management is only possible if the user is aware of the quantity of water he uses and the quantity available to him. Traditional water meters can't continuously monitor water usage and don't provide an efficient method of monitoring it.
- Conventional systems for monitoring water levels are either manual checking or simple float-based indicators, leading to time consumption, errors, and no real-time data. With respect to such limitations, a water tank monitoring system will be designed accordingly.
- A water tank monitoring system is used to sort out the issues associated with water tanks. It is also possible to check the level of the water, using a sensor and switch the pump or valve on or off based on the signal received from the microcontroller, using a relay module.
- The project aims to measure the water level in a tank, display the water level on the LCD 1602 display, and control the pump according to the water level.

- This project aims at the design, development, and implementation of an effective system capable of continuously monitoring water levels in tanks with perfect accuracy. The system combines the latest technologies in sensors, and microcontrollers to deliver real-time data on the water level for the effective management of available water resources. It contains an alarm to provide alerts.

Problem and motivation

❖ Problems that we are going to solve through this project.

- How to identify the current water level?
- How to control the water pump's working condition at the right time?
- How to use time and energy efficiently while minimizing human intervention?

❖ Benefits of this project

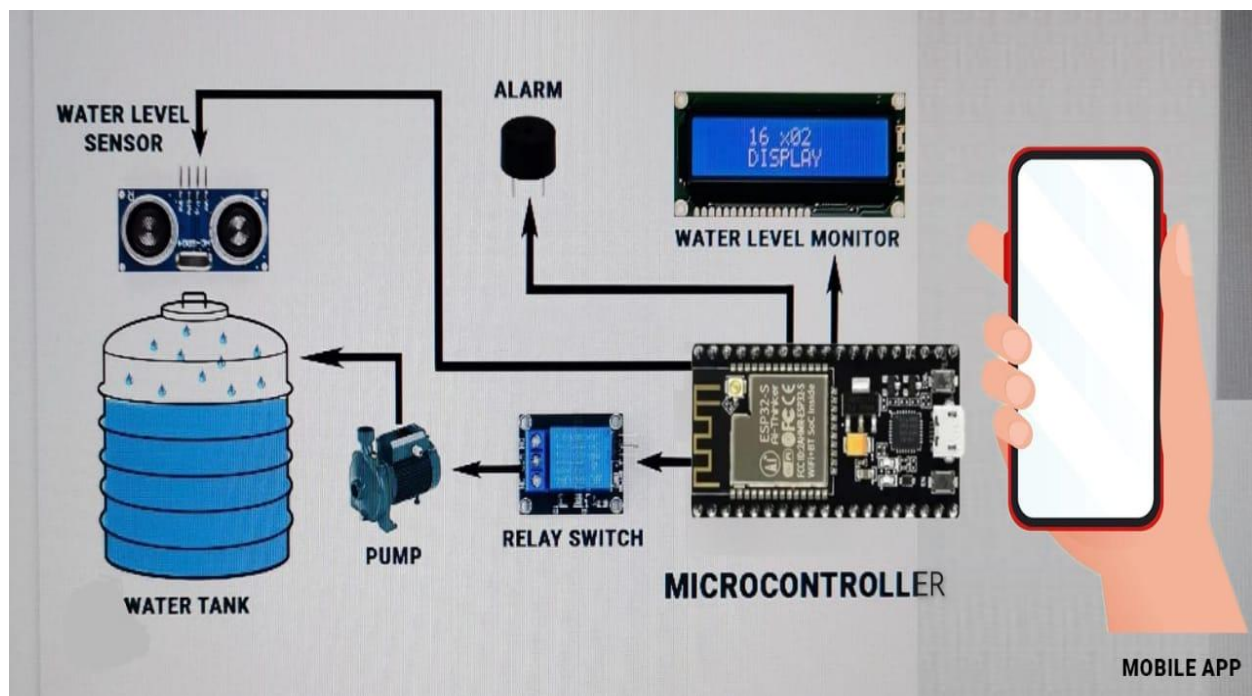
- Display the real-time water level
- Automatically pump up the water when the level is low
- Provide notification by alarm
- Control and monitor the system via mobile phone
- Minimizing human labor
- Reducing water waste
- Preventing water shortage

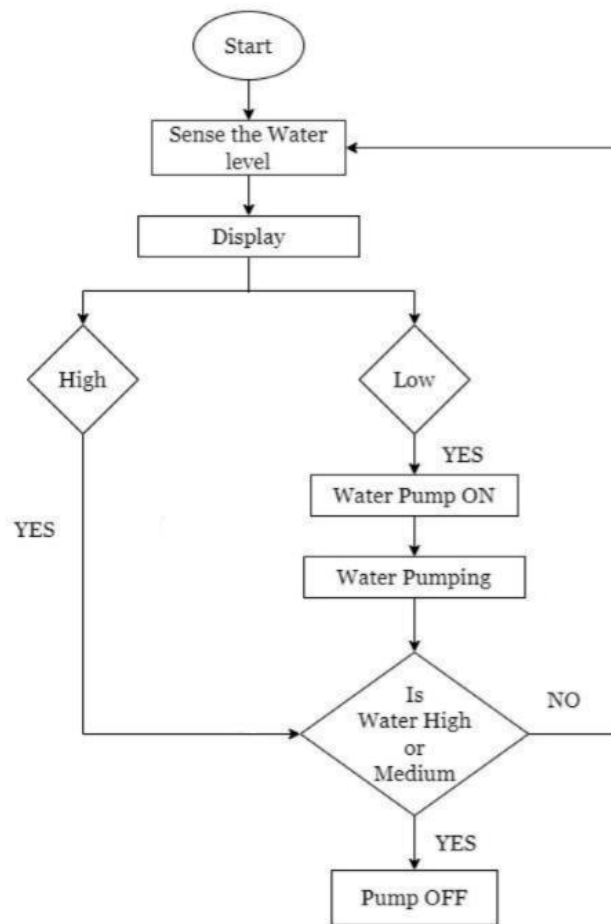
Aim and objectives

❖ Specific objectives of this project are:

- To develop a reliable water level monitoring system that will prove cost-effective.
- Provision of real-time water level data for effective decision-making.
- Contribution to the efforts of saving water through the prevention of wastage.
- Improved water management practices at residential, commercial, and industrial levels.
- Providing opportunity to live daily life more comfortably by reducing human labor.

System diagram





Methodology

❖ Used Components

- ESP8266 microcontroller
- Ultrasonic sensor (HC-SR04)
- Relay module
- LCD 1602 display
- Water Pump
- 5V Power Supply
- Alarm

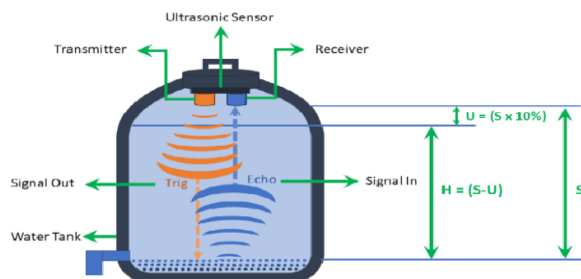
❖ Methods and Technologies

➤ Process the inputs and connect with the mobile app:

- Use an **ESP8266 microcontroller** as the brain of this circuit. This is a low cost WIFI microcontroller, with built in TCP/IP networking software, and microcontroller capability.

➤ Real-time water level monitoring:

- Use an **Ultrasonic sensor** to measure and detect water levels accurately and effectively.
- This sensor uses ultrasonic waves to identify the distance between the sensor and the water surface. According to the time, that waves take to travel to the surface of the water and back is used to calculate the water level.



➤ Display the water level:

- Use an **LCD 1602 display** to show the current water level as a percentage.

➤ Controls the pump or valve:

- A **Relay module** is used to control the water pump.
 - If the water level is lower than 20%, the relay is activated, turning the pump on and starting to fill the tank.
 - If the water level is higher than 95%, the relay is deactivated, turning the pump off to stop overflow.

- **Get an alarm notification:**
 - The **alarm** will notify you when the water level high and low.
- **Pump the water :**
 - Use a **water pump** to pump up the water to tank.
- **Provide power to the microcontroller**
 - For powerup the system, there is a **5v power supply**.

Evaluation Method

- Test the ultrasonic sensor to make sure that it accurately reads across different water levels
- Confirm the communication between the ultrasonic sensor, ESP 8266 module, and the LCD 1602 display
- Assess the system's power usage to ensure performance, especially for powered setups.
- Verify that the motor units seamlessly with the relay module and ESP8266.
- Ensure, the motor can be controlled with the signals from ESP 8266 and the relay module.
- Compile your code and upload it to the ESP8266. Track the code size to make sure it fits with the Flash memory limits.

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